

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df = pd.read_csv('/content/netflix.csv')
```

df



	show_id	type	title	director	cast	country	date_added	release_year
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	September 25, 2021	2020
1	s2	TV Show	Blood & Water	NaN	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	September 24, 2021	2021
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	NaN	September 24, 2021	2021
3	s4	TV Show	Jailbirds New Orleans	NaN	NaN	NaN	September 24, 2021	2021
4	s5	TV Show	Kota Factory	NaN	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	September 24, 2021	2021
...	...	...	...	...	...	...	...	...
8802	s8803	Movie	Zodiac	David Fincher	Mark Ruffalo, Jake Gyllenhaal, Robert Downey J...	United States	November 20, 2019	2007
8803	s8804	TV Show	Zombie Dumb	NaN	NaN	NaN	July 1, 2019	2018
8804	s8805	Movie	Zombieland	Ruben Fleischer	Jesse Eisenberg, Woody Harrelson,	United States	November 1, 2019	2009

Next steps:

Generate code with df

 View recommended plots

```
df.shape
```



```
(8807, 12)
```

```
df.sample(10)
```

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description
5506	s5507	Movie	Mostly Sunny	Dilip Mehta	Sunny Leone	Canada	May 1, 2017	2016	TV-MA	83 min	Documentaries, International Movies	Explore the unconventional career of Sunny Leone
751	s752	TV Show	Guru Aur Bhole	NaN	Amit Kumar	NaN	June 8, 2021	2018	TV-Y7	2 Seasons	Kids' TV, TV Comedies	A singer and dancer who are best friends use
5493	s5494	Movie	Handsome: A Netflix Mystery Movie	Jeff Garlin	Jeff Garlin, Natasha Lyonne, Amy Sedaris, Stev...	United States	May 5, 2017	2017	TV-MA	81 min	Comedies	Homicide detective Geri Handson sets out to solve
3295	s3296	TV Show	Green Eggs and Ham	NaN	Michael Douglas, Adam Devine, Keegan-Michael K...	United Kingdom, Canada, United States	November 8, 2019	2019	TV-Y7	1 Season	Kids' TV, TV Action & Adventure, TV Comedies	On a road trip to save a endangered animal, p
47	s48	TV Show	The Smart Money Woman	Bunmi Ajakaiye	Osas Ighodaro, Ini Dima-Okojie, Kemi Lala Akin...	NaN	September 16, 2021	2020	TV-MA	1 Season	International TV Shows, Romantic TV Shows, TV ...	Five glamorous millennial women strive for success
7399	s7400	Movie	Mansfield Park	Patricia Rozema	Frances O'Connor, Embeth Davidtz, Jonny Lee Mi...	United Kingdom	January 1, 2020	1999	PG-13	112 min	Comedies, Dramas, International Movies	A poor heroine is sent to live with wealthy relatives
6648	s6649	Movie	Dream Boat	Tristan Ferland Milewski	NaN	Germany	December 1, 2017	2017	TV-MA	92 min	Documentaries, International Movies, LGBTQ Movies	On a cruise ship, gay men, five passengers die
5827	s5828	TV Show	LEGO Bionicle: The Journey to One	NaN	Nolan Balzer, Paolo Bryant, Jacqui Fox, Quinn ...	United States	July 29, 2016	2016	TV-Y7	2 Seasons	Kids' TV	Six legendary heroes find themselves in an epic
1568	s1569	Movie	Kalel, 15	Jun Lana	Elijah Canlas, Eddie Garcia, Jaclyn Jose, Gabb...	Philippines	December 9, 2020	2019	TV-MA	105 min	Dramas, Independent Movies, International Movies	Surrounded by tensions and secrets, teenage
212	s213	TV	Rebellion	NaN	Charlie Murphy, Ruth Bradley	Ireland	August 27,	2016	TV-MA	2	International TV Shows, TV	As World War II rages, three

```
df.isnull().sum()
```

```

show_id      0
type         0
title        0
director    2634
cast        825
country     831
date_added   10
release_year  0
rating       4
duration     3
listed_in    0
description  0
dtype: int64

```

```
df_director_r = pd.DataFrame(df['director'].apply(lambda x:str(x).split(',')).tolist(), index=df['title'])
df_director = df_director_r.stack().reset_index()
df_director.drop('level_1', axis=1, inplace=True)
df_director.rename(columns={0:'director'}, inplace=True)
df_director.head(10)
```



	title	director
0	Dick Johnson Is Dead	Kirsten Johnson
1	Blood & Water	nan
2	Ganglands	Julien Leclercq
3	Jailbirds New Orleans	nan
4	Kota Factory	nan
5	Midnight Mass	Mike Flanagan
6	My Little Pony: A New Generation	Robert Cullen
7	My Little Pony: A New Generation	José Luis Ucha
8	Sankofa	Haile Gerima
9	The Great British Baking Show	Andy Devonshire



Next steps:

[Generate code with df_director](#)

 [View recommended plots](#)

```
df_cast_r = pd.DataFrame(df['cast'].apply(lambda x:str(x).split(',')).tolist(), index=df['title'])
df_cast = df_cast_r.stack().reset_index()
df_cast.drop('level_1', axis=1, inplace=True)
df_cast.rename(columns={0:'cast'}, inplace=True)
df_cast.head(10)
```



	title	cast
0	Dick Johnson Is Dead	nan
1	Blood & Water	Ama Qamata
2	Blood & Water	Khosi Ngema
3	Blood & Water	Gail Mabalane
4	Blood & Water	Thabang Molaba
5	Blood & Water	Dillon Windvogel
6	Blood & Water	Natasha Thahane
7	Blood & Water	Arno Greeff
8	Blood & Water	Xolile Tshabalala
9	Blood & Water	Getmore Sithole



Next steps:

[Generate code with df_cast](#)

 [View recommended plots](#)

```
df_new = df_director.merge(df_cast, how = 'inner', on = 'title')

df_new
```



	title	director	cast
0	Dick Johnson Is Dead	Kirsten Johnson	nan
1	Blood & Water	nan	Ama Qamata
2	Blood & Water	nan	Khosi Ngema
3	Blood & Water	nan	Gail Mabalane
4	Blood & Water	nan	Thabang Molaba
...	...	...	...
70807	Zubaan	Mozez Singh	Manish Chaudhary
70808	Zubaan	Mozez Singh	Meghna Malik
70809	Zubaan	Mozez Singh	Malkeet Rauni
70810	Zubaan	Mozez Singh	Anita Shabdish
70811	Zubaan	Mozez Singh	Chittaranjan Tripathy



70812 rows × 3 columns

Next steps:

[Generate code with df_new](#)[View recommended plots](#)

```
df.columns

Index(['show_id', 'type', 'title', 'director', 'cast', 'country', 'date_added',
      'release_year', 'rating', 'duration', 'listed_in', 'description'],
      dtype='object')

df_final = df_new.merge(df[['show_id', 'type', 'title', 'date_added', 'country',
      'release_year', 'rating', 'duration', 'listed_in', 'description']], how = 'inner', on = 'title')

df_final.sample(10)
```



	title	director	cast	show_id	type	date_added	country	release_yea
11592	Hello, Me!	nan	Kim Byung-choon	s1297	TV Show	February 18, 2021	South Korea	202
34192	Shonar Pahar	Parambrata Chatterjee	Tanuja	s4089	Movie	February 22, 2019	India	201
27787	Brother	Julien Abraham	Lisette Malidor	s3238	Movie	November 22, 2019	France	201
11418	The Conjuring	James Wan	Lili Taylor	s1284	Movie	February 21, 2021	United States	201
51498	Charlie and the Chocolate Factory	Tim Burton	Philip Wiegratz	s6447	Movie	January 1, 2020	United Kingdom, United States	200
13347	The Con Is On	James Haslam	Maggie Q	s1511	Movie	December 21, 2020	United States, United Kingdom	201
24647	Full Count	Robert Eagar	E. Roger Mitchell	s2888	Movie	February 23, 2020	United States	201
2169	Black Island	Miguel Alexandre	Sammy Scheuritzel	s253	Movie	August 18, 2021	NaN	202
46525	X: Past Is Present	Sudhish Kamath	Usha Uthup	s5838	Movie	July 1, 2016	India	201

```
df_final['cast'].replace(['nan'], ['unknown Actor'], inplace=True)
df_final['director'].replace(['nan'], ['unknown director'], inplace=True)
df_final.head()
```



	title	director	cast	show_id	type	date_added	country	release_year	rating
0	Dick Johnson Is Dead	Kirsten Johnson	unknown Actor	s1	Movie	September 25, 2021	United States	2020	PG-13
1	Blood & Water	unknown director	Ama Qamata	s2	TV Show	September 24, 2021	South Africa	2021	TV-M
2	Blood & Water	unknown director	Khosi Ngema	s2	TV Show	September 24, 2021	South Africa	2021	TV-M

Next steps:

[Generate code with df_final](#)
[View recommended plots](#)

```
df_final[df_final['duration'].isnull()]
df_final
```



	title	director	cast	show_id	type	date_added	country	release_year
0	Dick Johnson Is Dead	Kirsten Johnson	unknown Actor	s1	Movie	September 25, 2021	United States	2020
1	Blood & Water	unknown director	Ama Qamata	s2	TV Show	September 24, 2021	South Africa	2021
2	Blood & Water	unknown director	Khosi Ngema	s2	TV Show	September 24, 2021	South Africa	2021
3	Blood & Water	unknown director	Gail Mabalane	s2	TV Show	September 24, 2021	South Africa	2021
4	Blood & Water	unknown director	Thabang Molaba	s2	TV Show	September 24, 2021	South Africa	2021
...	...	...	...	...	...	...	...	...
70807	Zubaan	Mozez Singh	Manish Chaudhary	s8807	Movie	March 2, 2019	India	2015
70808	Zubaan	Mozez Singh	Meghna Malik	s8807	Movie	March 2, 2019	India	2015
70809	Zubaan	Mozez Singh	Malkeet Rauni	s8807	Movie	March 2, 2019	India	2015

Next steps:

[Generate code with df_final](#)
[View recommended plots](#)

1. Find the count of each categorical variable both using graphical and non-graphical analysis

a. For non_graphical Analysis

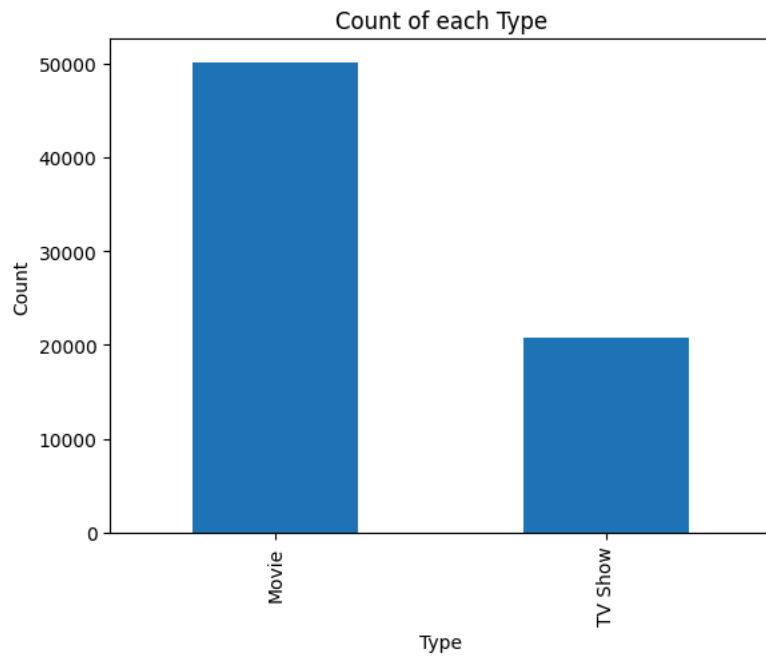
```
type_counts = df_final['type'].value_counts()
print(type_counts)
```



```
type
Movie      50098
TV Show    20714
Name: count, dtype: int64
```

b. For graphical Analysis

```
import matplotlib.pyplot as plt
type_counts.plot(kind='bar')
plt.title('Count of each Type')
plt.xlabel('Type')
plt.ylabel('Count')
plt.show()
```



2. Comparison of tv shows vs movies

a. Find the number of movies produced in each country and pick the top 10 countries

```
movies_df = df_final[df_final['type'] == 'Movie']
movies_count = movies_df.groupby('country')['title'].nunique()
movies_count.sort_values(ascending=False).head(10)
```



```
country
United States    2058
India            893
United Kingdom   206
Canada           122
Spain            97
Egypt            92
Nigeria          86
Indonesia        77
Turkey           76
Japan            76
Name: title, dtype: int64
```

b. Find the number of Tv_Shows produced in each country and pick the top 10 countries.

```
tv_show_df = df_final[df_final['type'] == 'TV Show']
tv_show_count = tv_show_df.groupby('country')['title'].nunique()
tv_show_count.sort_values(ascending=False).head(10)
```



```
country
United States    760
United Kingdom   213
Japan            169
South Korea      158
India            79
Taiwan           68
Canada           59
France           49
Australia        48
Spain            48
Name: title, dtype: int64
```

3. What is the best time to launch a TV show ?

a. Find which is the best week to release the Tv-show or the movie. Do the analysis separately for Tv-show and movies

```
df_final['date_added'] = pd.to_datetime(df_final['date_added'])[0])
df_final['week'] = df_final['date_added'].dt.isocalendar().week
movies_df = df_final[df_final['type'] == 'Movie']
movies_df.groupby('week')['type'].value_counts()
```

```
↩ week type
   38  Movie    50098
Name: count, dtype: int64
```

Ans - 38week 50098 total number of movies

```
df_final['date_added'] = pd.to_datetime(df_final['date_added'])[0])
df_final['week'] = df_final['date_added'].dt.isocalendar().week
tvshow_df = df_final[df_final['type'] == 'TV Show']
tvshow_df.groupby('week')['type'].value_counts()
```

```
↩ week type
   38  TV Show    20714
Name: count, dtype: int64
```

Ans - 38week 20714 total number of TV show

b.Find which is the best month to release the Tv-show or the movie. Do the analysis separately for Tv-shows and Movies

```
df_final['date_added'] = pd.to_datetime(df_final['date_added'])[0])
df_final['month'] = df_final['date_added'].dt.month
movies_df = df_final[df_final['type'] == 'Movie']
movies_df.groupby('month')['type'].value_counts()
```

```
↩ month type
   9      Movie    50098
Name: count, dtype: int64
```

Ans - 9 month 50098 total number of movies

```
df_final['date_added'] = pd.to_datetime(df_final['date_added'])[0])
df_final['month'] = df_final['date_added'].dt.month
tvshow_df = df_final[df_final['type'] == 'TV Show']
tvshow_df.groupby('week')['type'].value_counts()
```

```
↩ week type
   38  TV Show    20714
Name: count, dtype: int64
```

9 month 20174 total number of TV show

4. Analysis of actor/directors of different types of shows/movies.

a.identify the top 10 director who have appeared in most movies or TV shows.

```
tv_show_df = df_final[df_final['type'] == 'TV Show']
```

```
unique_title_counts = tv_show_df.groupby('cast')['title'].nunique().reset_index()
unique_title_counts.columns = ['cast', 'unique_title_counts']
unique_title_counts.sort_values(by='unique_title_counts', ascending=False).head(10)
```




	cast	unique_title_counts	
15496	unknown Actor	350	
11974	Takahiro Sakurai	24	
13204	Yuki Kaji	17	
6136	Junichi Suwabe	17	
222	Ai Kayano	17	
2573	Daiki Yamashita	14	

b.identify the top 10 director who have appered in most movies or Tv Shows

```
tv_movie_df = df_final[df_final['type'] == 'Movie']
unique_title_counts = tv_movie_df.groupby('director')['title'].nunique().reset_index()
unique_title_counts.columns = ['director', 'unique_title_counts']
unique_title_counts.sort_values(by='unique_title_counts', ascending=False).head(10)
```



	director	unique_title_counts	
4874	unknown director	188	
3818	Rajiv Chilaka	22	
234	Jan Suter	18	
3865	Raúl Campos	18	
4429	Suhas Kadav	16	
2308	Jay Karas	15	
3067	Marcus Raboy	15	
1289	Cathy Garcia-Molina	13	
3135	Martin Scorsese	12	